

Question Paper

Exam Date & Time: 13-May-2025 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Examination - April / May -2025

ADVANCED DATA STRUCTURE [IT-5013]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

1) Consider the following pseudo-code: (4)

A)

```
def func(n):  
    for i in range(n):          # Loop A  
        j = 1  
        while j < n:           # Loop B  
            j *= 2
```

- (i) **Analyze** the overall time complexity of the above function using asymptotic notation.
- (ii) Clearly distinguish the role of each loop and how they contribute to the final complexity.

Provide justification for why the inner loop behaves logarithmically.

Course Outcome: "CO1" / Cognitive Level: "Application"

B) A dynamic array doubles its size whenever its capacity is full. (4)

- i. **Analyze** the time complexity of inserting n elements into such an array using **amortized analysis**.
- ii. **Compare** this with the **worst-case time complexity** of a single insertion when resizing occurs.

Draw a conclusion about why amortized analysis is more informative in this scenario.

Course Outcome: "CO1" / Cognitive Level: "Analysis"

C) **Provide a real-world scenario** that can be modeled as an NP-Complete problem. **Identify** the name of the problem and the challenge it presents. (2)

Course Outcome: "CO1" / Cognitive Level: "Application"

2) You are given the following list of integers representing task priorities: (4)
[15, 7, 20, 3, 10, 8, 12, 5, 18, 6]

A)

1. **Construct a min-heap** from this array using the given order of insertion.
2. Then, **insert three new priority values** one by one: 2, 1, 9

After each insertion, **update** the array representation of the heap.

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) **Insert the following keys into a hash table of size 11** (3)
using hash function:

$$h(k) = k \bmod 11$$

Keys: [23, 44, 39, 15, 28, 20, 33, 51]

Use quadratic probing for collision resolution and illustrate the final hash table.

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) A real-time system must handle tasks with different priority levels (lower number = higher priority). (3)
Apply a data structure that schedules and executes the highest-priority task first.
Apply your structure to insert the following tasks with priorities:
 {(TaskA, 3), (TaskB, 1), (TaskC, 2), (TaskD, 1)}
Show the order in which tasks will be executed.

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) **Construct** a 2-3 Tree by **inserting** the following sequence of keys: (4)
 [5, 10, 15, 20, 25, 30, 35, 40, 45, 50]
 A) Solve the problem step-by-step and **illustrate** the tree after all insertions.
 Explain how the **2-3 Tree properties** (nodes with 2 or 3 children, sorted values) are preserved.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) You are given the input values to construct a binary tree (not necessarily a BST): (3)
 [A, B, C, D, E, F, G]
 Assume the tree is built in level-order. Apply post-order traversal on the tree and write the result.

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) **Construct** a Binary Search Tree by inserting the following sequence of keys: (3)
 [45, 30, 60, 20, 50, 70, 10]
Demonstrate the final tree structure and **verify** that it satisfies the BST property.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 4) Given the array: (4)
 [32, 15, 7, 89, 42, 12, 64, 25, 19, 5, 37]
 A) Apply the Quick Sort algorithm using the first element as the pivot.
 Show the recursive partitioning steps and the final sorted array.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) You are given the array: (3)
 [2, -4, 3, -1, 2, -5, 4, 6, -3, 2]
 Apply the divide-and-conquer method to solve the maximum subarray sum problem.

Break down the array into subproblems and show how the solution is built.

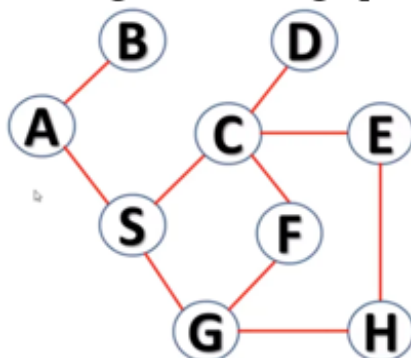
Course Outcome: "CO4" / Cognitive Level: "Application"

- C) You are given the following characters and their frequencies: (3)
 A (5), B (9), C (12), D (13), E (16), F (45)
 Apply the greedy Huffman encoding strategy to build the optimal prefix code.
 Show the process of tree construction and the final code for each character.

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- 5) Consider the following undirected graph: (4)

A)



Apply BFS starting from vertex A.
 List the order of nodes visited and show the contents of the queue at each step.

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Evaluate the suitability of Dijkstra's Algorithm and Bellman-Ford Algorithm for finding the shortest path in a weighted graph with negative edge weights. (4)
 Compare their time complexities and justify which algorithm is more appropriate under different conditions.

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

- C) Identify two scenarios where using 2-3-Trees could be more beneficial than AVL trees. (2)
 Mention the trade-offs in terms of rotation cost and complexity.

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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MIT MPL and BLR- BTech-M Sc-MCA - II-IV and VI Sem - End Semester Examination -
 Apr-May 2025

IT-5013:ADVANCED DATA
 STRUCTURE

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* NA - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	9	28(56%)
4	Analysis	4	15(30%)
5	Create	2	7(14%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)